

GAIA Statement: I **did** ☒ / **did not** ☐ use a form of generative AI during the completion of this assignment. If I did, I understand that I must also turn in a completed copy of the course's GAIA Disclosure Worksheet for this assignment.

BE 493: Biomedical Measurements and Analysis

Characterization of Particle Size and Polydispersity in Greek and Regular Yogurt via Dynamic Light Scattering

Junaid Ali, Mark Cho, Ashley Dhar, Jaime Gonzalez-Munoz, Aliza Monaldo

Boston University
Department of Biomedical Engineering
Spring 2026

1. Introduction

1.1. Theory

Dynamic light scattering (DLS) is a non-destructive technique used to measure the hydrodynamic size of particles in suspension. The technique relies on analyzing temporal fluctuations in the intensity of scattered laser light as particles undergo Brownian motion within a liquid sample.^{1,2} More simply, when a laser beam hits the suspension, the moving particles create interference and cause the beam to scatter over time, directly affecting the intensity. These fluctuations are quantified by the intensity autocorrelation function $g^2(\tau)$, which for a monodisperse suspension decays exponentially:¹

$$g^2(\tau) = 1 + \beta \cdot \exp(-2\Gamma\tau) \quad (1)$$

where τ is the delay time (lag time) between intensity measurements, β is a coherence factor and $\Gamma = D_t \cdot q^2$ is the decay rate.^{1,2} D_t is the translational diffusion coefficient: smaller particles diffuse faster and produce faster decays; larger particles produce slower ones. This makes Γ a direct measure of particle size.^{1,2} $q = (4\pi n/\lambda_0) \sin(\theta/2)$ is the scattering vector (the momentum transfer between incident and scattered photons, defined as $\mathbf{q} = \mathbf{k}_s - \mathbf{k}_i$) with magnitude set by the solvent refractive index n , laser wavelength λ_0 , and detection angle θ .^{1,2} The hydrodynamic diameter d_H is then obtained from D_t via the Stokes–Einstein relation:^{1,2}

$$d_H = \frac{k_B T}{3\pi\eta D_t} \quad (2)$$

where k_B is the Boltzmann constant, T is absolute temperature, and η is the solvent viscosity. Together, Equations 1 and 2 form the core measurement chain: the autocorrelation decay yields D_t , and the Stokes–Einstein relation converts D_t to d_H .

Figure 1 illustrates how $g^2(\tau)$ changes in shape as sample complexity increases, ranging from the ideal monodisperse case to conditions representative of this experiment.

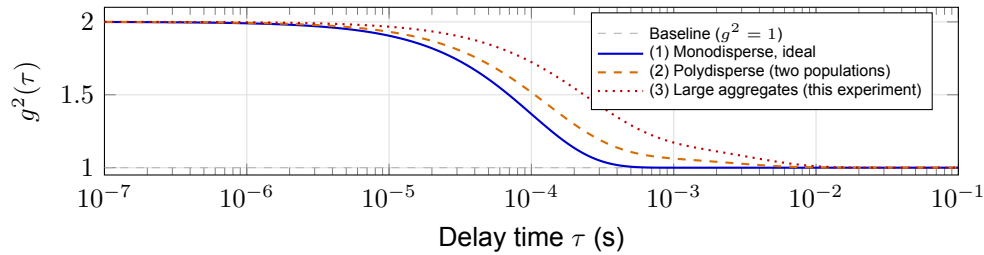


Figure 1: $g^2(\tau)$ for increasing levels of physical complexity. (1) An ideal monodisperse suspension yields a clean single-exponential decay at short lag times. (2) A polydisperse sample with two distinct size populations produces a multi-component decay spread over a wider time range. (3) A system dominated by large aggregates, representative of this experiment, decays slowly and broadly, consistent with the measured autocorrelation curves shown in Figure 2.

For polydisperse samples, cumulants analysis (a statistical method used to describe the shape of a probability distribution) yields two key outputs: the Z-average diameter (intensity-weighted mean d_H) and the polydispersity index (PDI), a dimensionless measure of size distribution breadth (0 = perfectly monodisperse; ≥ 0.7 = highly polydisperse).^{2,3} Because yogurt contains casein micelles (100–300 nm), whey proteins, and fat globules, both the Z-average diameter and PDI are needed to characterize the colloidal composition of the sample.⁴

1.2. Practice

The Anton Paar Litesizer 500 measures particle sizes ranging from 0.3 nm to 10 μm using a 658 nm semiconductor laser and avalanche photodiode (APD) detectors positioned at 15°, 90°,

and 175° .² We used backscatter detection at 175° for all measurements, as this geometry minimizes contributions from large aggregates and dust and reduces multiple-scattering artifacts in turbid samples.^{1,3} Scattered photons collected by the APD are converted to electrical pulses and processed by a digital correlator, which computes $g^2(\tau)$ across lag times spanning nanoseconds to seconds. The Kalliope software then applies cumulant analysis to report Z-average diameter and PDI. The diffusion coefficient D_t extracted from the autocorrelation decay is passed directly into Equation 2, the Stokes–Einstein relation, using the temperature and viscosity set in Kalliope, to convert the APD signal into a physical particle diameter.

To illustrate an ideal measurement, consider a NIST-traceable suspension of 100 nm polystyrene beads in water at 25°C (a certified reference material of known diameter used to verify instrument accuracy).⁵ Such a sample is nearly monodisperse ($\text{PDI} < 0.05$), so the APD produces a rapidly decorrelating $g^2(\tau)$ that fits cleanly to a single exponential, yielding a Z-average diameter close to 100 nm with low uncertainty. This ideal response, shown as curve (1) in Figure 1, contrasts sharply with the multi-component, slowly decaying autocorrelation functions produced by the yogurt samples in this experiment (curve (3)), where protein aggregates at 3000–4000 nm dominate the signal.

To perform a measurement, we load a prepared sample into a cuvette and place it in the temperature-controlled measurement chamber. In Kalliope, we set the solvent type, material type, measurement temperature, and scattering angle; these selections determine the viscosity and refractive index values used in the Stokes–Einstein calculation, making accurate parameter input critical for reliable size results. The software averages the autocorrelation functions across all acquisition runs, applies cumulant analysis, and flags any runs with poor correlation quality. We typically perform a blank measurement prior to sample analysis to confirm the absence of contaminants or dust.

1.3. Use Cases

DLS has been widely used to characterize casein micelles in dairy samples. Dalgleish and Hallett⁶ established DLS as a quantitative tool for dairy colloids, demonstrating that Z-average diameters of 150–200 nm could be reproducibly obtained from diluted milk across a range of dilution concentrations. This foundational work validated cumulant outputs as reliable metrics for dairy particle sizing and forms the measurement basis for the present experiment.

Building on this, Glantz et al.⁷ applied DLS to measure casein micelle sizes across milks from individual cows, reporting hydrodynamic diameters of 140–220 nm and PDI values of 0.15–0.37. Their study further showed that smaller casein micelles correlate with stronger gel formation, establishing that micelle size has direct functional consequences for fermented dairy products such as yogurt.

Thirdly, Beliciu and Moraru⁸ used DLS to investigate how solvent choice and measurement temperature affect the accuracy of casein micelle sizing, with the goal of establishing a validated measurement protocol. They performed cumulant analysis on raw and pasteurized skim milk diluted in three solvents (e.g. simulated milk ultrafiltrate, ultrafiltered water, and ultrafiltered permeate) at 6, 20, and 50°C using a 90Plus Nanoparticle Size Analyzer (Brookhaven Instruments). One-way ANOVA confirmed that both solvent and temperature had a statistically significant effect on the measured Z-average diameter. The authors concluded that ultrafiltered permeate is the most suitable solvent for DLS measurement of casein micelles, and that both solvent and temperature must be carefully controlled to obtain reproducible size data.

In the context of our study, Greek yogurt is produced by straining regular yogurt to remove acid whey, thereby concentrating the protein-rich curd phase. Ntemiri et al.⁹ showed that this process substantially increases total protein and casein content relative to regular yogurt.^{4,9} Because

Greek yogurt retains a higher casein-to-whey ratio and greater overall protein concentration,^{4,9} differences in DLS-measured particle size and PDI between the two yogurt types are mechanistically plausible: concentrated casein networks and altered micelle aggregation states during acidification may shift both the mean size and the breadth of the distribution.^{7,10} We apply DLS to test that hypothesis, comparing Z-average diameter and PDI between Greek and regular yogurt using a one-way analysis of variance.

2. Materials and Methods

2.1. Sample Preparation

We prepared Dannon vanilla whole milk yogurt (regular) and Dannon Light+Fit vanilla Greek yogurt for DLS measurement by diluting each to a 1:49 yogurt-to-deionized water ratio (w/w). For each sample, we weighed 1.0 g of yogurt on a tared analytical scale into a clean beaker and added 49.0 g of deionized water. We placed each beaker on a hotplate with a magnetic stir bar and mixed for 5–10 minutes until the suspension appeared visually homogeneous. We then loaded each suspension into three separate cuvettes, yielding three independent replicates per yogurt type (six cuvettes total). If a cuvette showed visible clumping or the instrument flagged poor signal quality, we remixed the sample and transferred it to a new cuvette before measurement.

2.2. Instrument Settings

We performed all DLS measurements on the Anton Paar Litesizer 500 using backscatter detection at 175°. We set the measurement temperature to 70°C, as elevated temperature increases particle mobility and reduces sample viscosity, promoting more consistent diffusion behavior during measurement. We set the solvent to water and the material type to protein, which determines the viscosity and refractive index values used in the Stokes–Einstein size calculation. The instrument had been previously calibrated; we did not perform a blank measurement prior to the yogurt trials. We recorded three measurements per yogurt type, corresponding to the three independent cuvettes per group.

2.3. Statistical Design

We performed a one-way analysis of variance (ANOVA) at a significance level of $\alpha = 0.05$ to test whether mean Z-average diameter and mean PDI differed significantly between Greek and regular yogurt. We selected one-way ANOVA as the most appropriate test because there is one categorical independent variable (yogurt type, two levels) and a continuous dependent variable (Z-average diameter or PDI), with independent replicates across groups. This approach has previously been applied to compare particle size distributions across colloidal systems, including nanostructured lipid carriers.¹¹ We performed all statistical calculations in MATLAB.

The ANOVA test statistic is:

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}} \quad (3)$$

where MS_{between} is the mean square between groups and MS_{within} is the mean square within groups, computed as¹²:

$$MS_{\text{between}} = \frac{SS_{\text{between}}}{k - 1}, \quad MS_{\text{within}} = \frac{SS_{\text{within}}}{N - k} \quad (4)$$

SS_{between} measures variability between group means, while SS_{within} measures variability within each group relative to its own mean.¹² A larger F -value indicates greater separation between group means relative to within-group variability.

3. Results

3.1. Data Summary

We used DLS measurements to quantify Z-average diameter and PDI for Greek and regular yogurt across three trials each. Greek yogurt exhibited a mean Z-average diameter of 3231 ± 375 nm, while regular yogurt exhibited a mean diameter of 4011 ± 1482 nm. Figure 2 shows the trial-averaged intensity autocorrelation functions for both groups. Both curves decay from a plateau near $g^2(\tau) \approx 1.8$ to the baseline ($g^2 = 1$) between approximately 10^{-4} and 10^{-2} s, indicating that particles of both types diffuse on similar timescales.

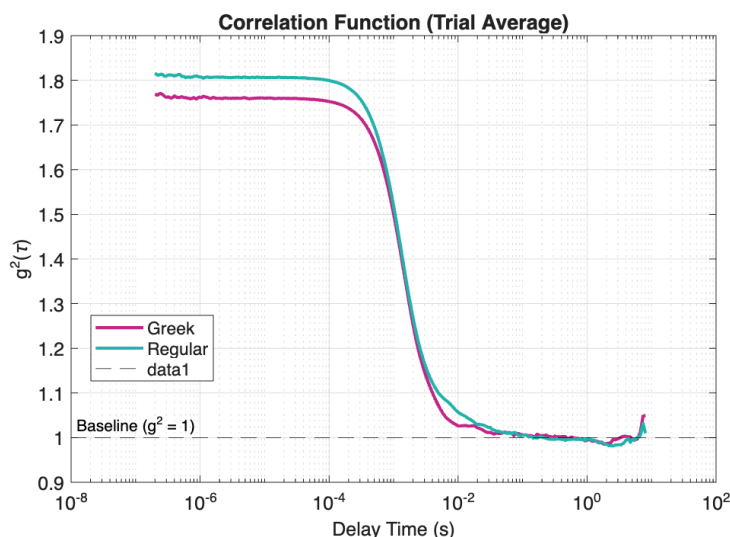


Figure 2: Trial-averaged intensity autocorrelation functions $g^2(\tau)$ for Greek (magenta) and regular (teal) yogurt measured by DLS at 175° backscatter. Both curves decay from a plateau near $g^2(\tau) \approx 1.8$ to the baseline ($g^2 = 1$, dashed) between $\sim 10^{-4}$ and 10^{-2} s. The dashed line denotes the theoretical baseline for a fully decorrelated signal.

Greek yogurt displayed a narrower, more consistent intensity-weighted size distribution than regular yogurt, as shown in Figure 3. The higher standard deviation in the regular yogurt group reflects greater trial-to-trial variability, driven in large part by the anomalous Trial 2 result discussed in Section 3.3.

Greek yogurt had a mean PDI of 0.45 ± 0.42 , indicating a moderately broad size distribution. Regular yogurt had a mean PDI of 3.56 ± 5.70 , driven largely by one anomalous trial (see Section 3.3). Table 1 reports all individual trial results. Figure 4 displays mean particle size and PDI by group with individual trials and the flagged outlier annotated.

Table 1: Z-average diameter and PDI for each trial and group mean. The flagged trial (Regular Trial 2) produced a PDI exceeding the physically valid range of 0–1.

Yogurt Type	Trial	Diameter (nm)	PDI
Greek	1	3456	0.243
Greek	2	3440	0.930
Greek	3	2798	0.176
Greek	Average	3231	0.450
Regular	1	2394	0.228
Regular	2	5303	10.1 [†]
Regular	3	4337	0.304
Regular	Average	4011	3.56

[†] Flagged: PDI > 1 is physically meaningless for DLS (valid range 0–1).

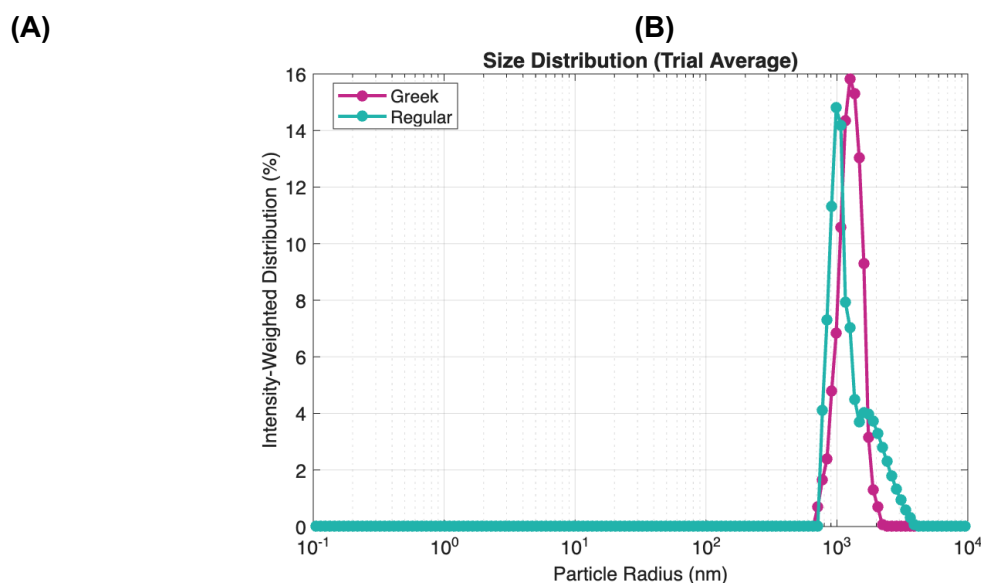


Figure 3: Trial-averaged intensity-weighted particle size distributions for Greek (magenta) and regular (teal) yogurt, derived from the autocorrelation functions in Figure 2 via cumulant analysis. Greek yogurt displays a narrower, more consistent distribution; regular yogurt shows a broader spread with greater trial-to-trial variability.

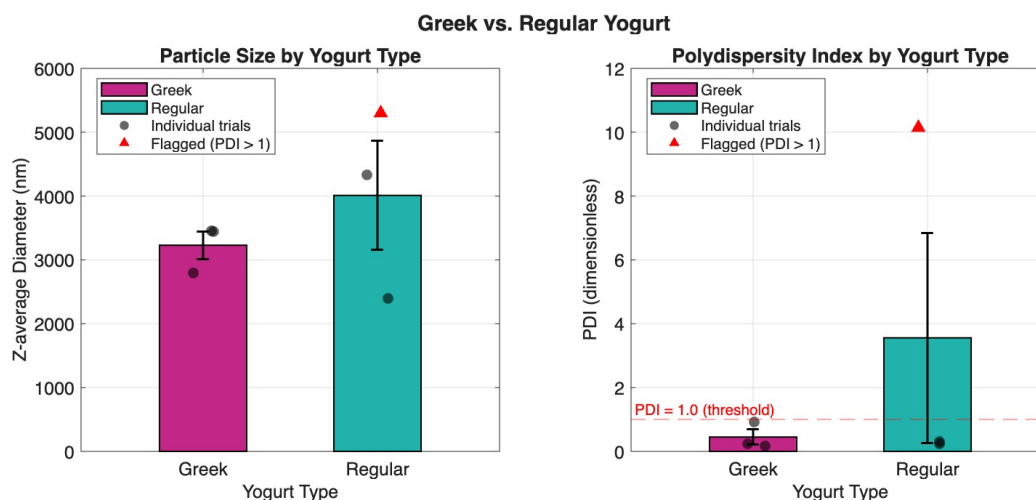


Figure 4: (A) Z-average particle diameter and (B) polydispersity index (PDI) for Greek and regular yogurt. Bars show group means; error bars indicate ± 1 standard error. Individual trials are shown as circles; the flagged trial (PDI > 1) is marked with a triangle. The red dashed line in (B) at PDI = 1.0 denotes the upper boundary of the physically valid DLS range.

3.2. Statistical Findings

We used a one-way ANOVA to test whether mean Z-average diameter and mean PDI differed significantly between Greek and regular yogurt ($\alpha = 0.05$). With two groups and three replicates per group, both tests had 1 degree of freedom between groups and 4 degrees of freedom within groups.

For Z-average diameter, the result was not statistically significant: $F(1,4) = 0.78$, $p = 0.43$. For PDI, the result was likewise not significant: $F(1,4) = 0.89$, $p = 0.40$. In both cases, we failed to reject the null hypothesis that Greek and regular yogurt have the same mean value.

3.3. Anomalies

Regular yogurt Trial 2 produced a PDI of 10.14, which falls well outside the physically valid range for DLS measurements (0–1). A PDI exceeding 1 indicates that the autocorrelation function

could not be reliably fit by the cumulant model, most likely due to large aggregates dominating the scattering signal, sample contamination, or insufficient homogenization at the time of measurement. This trial substantially inflated both the mean and variance of the regular yogurt PDI and contributed to the large standard deviation in diameter as well. Although we flagged this trial, we retained it in all analyses, which increased within-group variance and reduced statistical power. The non-significant ANOVA results therefore do not necessarily indicate that no true difference exists between the yogurt types; rather, they reflect the limited ability to resolve any potential difference given the high variability and small sample size.

4. Discussion

4.1. Inconsistencies

The measured Z-average diameters (3000–4000 nm) were substantially larger than the expected casein micelle size range of 150–300 nm.^{6,7} The instrument most likely detected large protein aggregates rather than individual casein micelles. Two experimental conditions contributed to this outcome: the 1:49 dilution was likely insufficiently dilute, allowing multiple scattering to bias the measured size upward, and the 70°C measurement temperature likely promoted whey protein denaturation and heat-induced association with casein micelles, both of which are known to drive aggregation.^{10,13} The anomalously high PDI in Regular Trial 2 is consistent with this interpretation, as large and poorly dispersed aggregates can cause the cumulant fit to fail entirely. Despite these issues, Greek yogurt showed a slightly lower mean diameter and more consistent PDI across trials than regular yogurt, which broadly aligns with the expectation that straining removes whey and produces a more uniform protein population.⁹

4.2. Limitations

A fundamental challenge with DLS is that scattering intensity scales with the sixth power of particle diameter, meaning even a small number of large aggregates can dominate the signal and mask the particles of interest. This size bias is particularly problematic in a concentrated matrix like yogurt, where hotplate mixing cannot guarantee the same degree of homogenization as high-shear or ultrasonic methods. Additionally, three replicates per group represents the minimum sample size for ANOVA. These replicates yield only four total degrees of freedom within groups, making results highly sensitive to outliers and substantially reducing the power of the statistical test. A minimum of five replicates per group would improve the reliability of our analysis.

4.3. Future Improvements

We can directly address the limitations described above with modifications to the sample preparation protocol. Filtering samples through a 0.45 μm syringe filter prior to measurement would remove aggregates while retaining individual casein micelles (100–300 nm). We should also increase the dilution from 1:49 to at least 1:99 to reduce multiple scattering, and lower the measurement temperature from 70°C to 25°C to prevent thermally-induced aggregation. Including a PDI acceptance criterion of ≤ 0.5 would provide an objective gate for flagging and rerunning unreliable measurements. Figure 5 summarizes this proposed protocol.

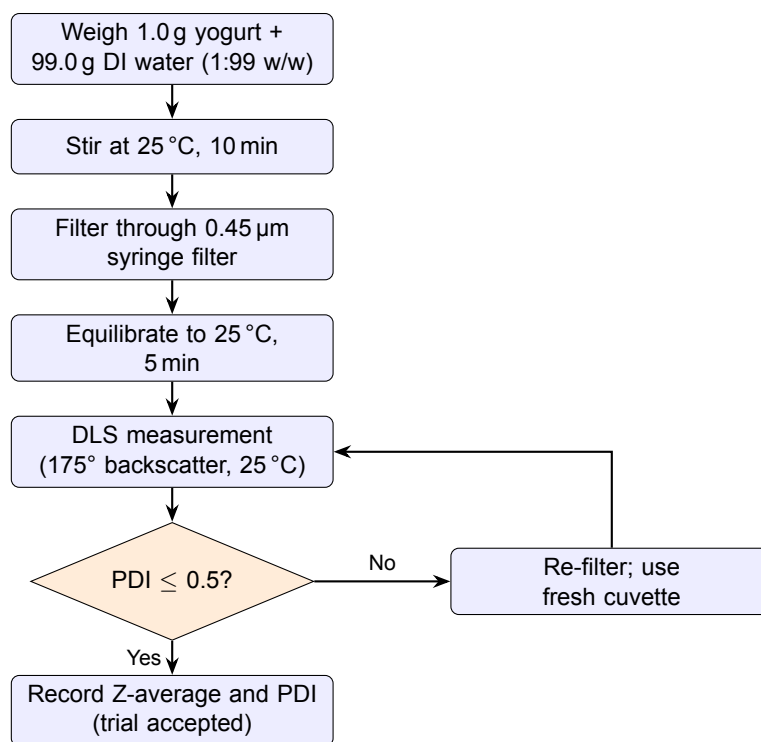


Figure 5: Proposed improved DLS protocol for yogurt characterization. A 1:99 dilution reduces multiple scattering; 0.45 μm filtration removes aggregates while retaining casein micelles (100–300 nm); measurement at 25 °C prevents thermally-induced aggregation. The PDI acceptance gate provides an objective criterion for rejecting and rerunning unreliable measurements.

4.4. Experiment Assessment

The Litesizer 500 is well suited to this type of measurement: its 175° backscatter geometry reduces aggregate interference relative to forward scatter, its measurement range of 0.3 nm to 10 μm covers the full relevant size range, and Kalliope automatically flags poor-quality autocorrelation fits. The limitations we encountered stemmed from the sample preparation protocol rather than from the instrument itself. The combination of insufficient dilution and elevated measurement temperature caused the instrument to detect aggregates rather than individual casein micelles. The revised protocol in Figure 5 would address both issues. From a statistical standpoint, a Welch’s t-test would be more appropriate than one-way ANOVA for future comparisons, as it does not assume equal variance between groups, which is an assumption that was clearly violated here given the high variability in the regular yogurt group.

A novel extension of this work would be to use the Litesizer 500 to perform time-resolved DLS measurements during active yogurt fermentation (taking sequential readings every five minutes as pH drops) to directly track casein micelle aggregation kinetics in real time. To our knowledge, this specific application of DLS to monitor the yogurt gelation process continuously, rather than as a single endpoint measurement, has not been published for Greek versus regular yogurt matrices. Such a time-series approach would allow the instrument’s autocorrelation output to serve as a real-time gelation sensor: as the casein network coarsens, the Z-average diameter would increase monotonically and PDI would broaden in a measurable, reproducible pattern. This would transform the Litesizer from a static characterization tool into a dynamic process monitor, with direct relevance to industrial quality control of fermented dairy products.

References

1. Berne, B. J., & Pecora, R. (1976). *Dynamic light scattering: With applications to chemistry, biology, and physics*. Wiley. <https://doi.org/10.1038/263357a0>
2. Anton Paar GmbH. (2018). *Litesizer 500 instruction manual*. Anton Paar. <https://www.anton-paar.com/us-en/products/litesizer-500/>
3. Stetefeld, J., McKenna, S. A., & Patel, T. R. (2016). Dynamic light scattering: A practical guide and applications in biomedical sciences. *Biophysical Reviews*, 8(4), 409–427. <https://doi.org/10.1007/s12551-016-0218-6>
4. Lee, W. J., & Lucey, J. A. (2010). Formation and physical properties of yogurt. *Asian-Australasian Journal of Animal Sciences*, 23(9), 1127–1136. <https://doi.org/10.5713/ajas.2010.r.05>
5. Hackley, V. A., & Clogston, J. D. (2015). *Measuring the hydrodynamic size of nanoparticles in aqueous media using batch-mode dynamic light scattering* (NIST Special Publication 1200-6). National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.SP.1200-6>
6. Dalgleish, D. G., & Hallett, F. R. (1995). Dynamic light scattering: Applications to food systems. *Food Research International*, 28(3), 181–193. [https://doi.org/10.1016/0963-9969\(94\)00053-B](https://doi.org/10.1016/0963-9969(94)00053-B)
7. Glantz, M., Devold, T. G., Vegarud, G. E., Lindmark Månsson, H., Stålhammar, H., & Paulsson, M. (2010). Importance of casein micelle size and milk composition for milk gelation. *Journal of Dairy Science*, 93(4), 1444–1451. <https://doi.org/10.3168/jds.2009-2856>
8. Beliciu, C. M., & Moraru, C. I. (2009). Effect of solvent and temperature on the size distribution of casein micelles measured by dynamic light scattering. *Journal of Dairy Science*, 92(5), 1829–1839. <https://doi.org/10.3168/jds.2008-1467>; <https://pubmed.ncbi.nlm.nih.gov/19389940/>
9. Ntemiri, K., Karastamatis, S., Zoidou, E., Moatsou, G., & Moschopoulou, E. (2022). Effect of modified manufacturing conditions on the composition of Greek strained yogurt and the quantity and composition of generated acid whey. *Foods*, 11(24), 3953. <https://doi.org/10.3390/foods11243953>
10. Mottar, J., Bassier, A., Joniau, M., & Baert, J. (1989). Effect of heat-induced association of whey proteins and casein micelles on yogurt texture. *Journal of Dairy Science*, 72(9), 2247–2256. [https://doi.org/10.3168/jds.s0022-0302\(89\)79355-3](https://doi.org/10.3168/jds.s0022-0302(89)79355-3)
11. Athiyyah, W., Soeratri, W., Rosita, N., & Hamdan, S. (2025). Optimization using D-optimal design of nanostructured lipid carrier (NLC) with variation of surfactants and co-surfactant. *Jurnal Farmasi dan Ilmu Kefarmasian Indonesia*, 12(1), 1–14. <https://doi.org/10.20473/jfiki.v12i12025.1-14>
12. Kim, T. K. (2017). Understanding one-way ANOVA using conceptual figures. *Korean Journal of Anesthesiology*, 70(1), 22–26. <https://doi.org/10.4097/kjae.2017.70.1.22>
13. Gastaldi, E., Lagaude, A., & Tarodo de la Fuente, B. (1996). Micellar transition state in casein between pH 5.5 and 5.0. *Journal of Food Science*, 61(1), 59–64. <https://doi.org/10.1111/j.1365-2621.1996.tb14725.x>